Building blocks

Build the three independent pieces every client and the pipeline depend on: validation, retry policy and the SQLite schema.

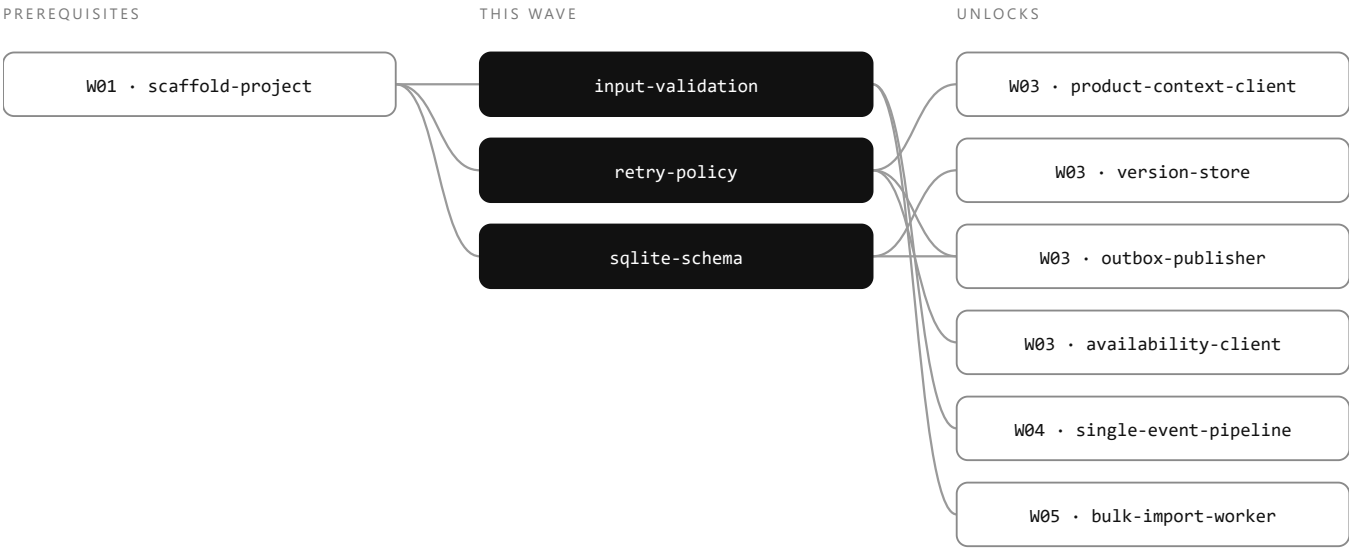
3 tasks | Requires Wave 01 | Unlocks Wave 03, 04, 05

Why this wave exists

These three have no dependency on each other, so they can be built in parallel. They are pure logic or storage with no network calls.

How it fits

The diagram shows this wave's tasks, the earlier-wave tasks they depend on, and the later-wave tasks that depend on them.



Hand-off. Wave 3 uses the retry helper in both HTTP clients, the database opener, the version and outbox tables and the lease table.

Tasks

Implementing input validation

`input-validation`

Depends on: scaffold-project

Implement pure validation functions with no I/O. validateSellerInventoryChanged checks required fields, countryCode as two uppercase letters, availableQuantity as an integer of at least 0 and sourceVersion as an integer. validateCsvHeader enforces the exact header order. validateCsvRow returns errors with the line number. Functions return result objects and never throw.

`src/domain/validation.ts · test/validation.test.ts`

- [ ] Field-level errors include the field name and a reason.
- [ ] A header mismatch is rejected before any row is processed.
- [ ] The eventType and schemaVersion values match `wiremock/__files/sample-event.json` .

Implementing retry policy

`retry-policy`

Depends on: scaffold-project

Create a generic retry helper with exponential backoff and full jitter. Max attempts, base delay and max delay come from config (RETRY_MAX_ATTEMPTS, RETRY_BASE_MS, RETRY_MAX_MS). Classify errors as retryable (429, 503, network errors and timeouts) or terminal (404, other 4xx and domain rejections). For 429 honour the Retry-After header in seconds and wait at least that long. Make the sleep function injectable. The numeric defaults are engineering choices, not values from the assignment, and must be justified in docs/decisions.md.

`src/infra/retry.ts · test/retry.test.ts`

- [ ] Retry-After: 5 produces a delay of at least 5 seconds, verified with an injected clock.
- [ ] Terminal errors are never retried.
- [ ] After the maximum attempts the last error is returned as a failure result, not thrown.

Creating SQLite schema and leases

`sqlite-schema`

Depends on: scaffold-project

Create the schema and a database opener. Enable WAL mode and foreign keys. Tables: stock_versions (composite key seller_id, product_id, size_id, country_code, plus applied_source_version), import_jobs (job_id, state, counters, timestamps), import_rows (job_id, line_number, outcome, reason), outbox (autoincrement id, unique event_id, stock_key, payload_json, published_at, attempts, last_error) and stock_leases (stock_key primary key, owner, expires_at_ms) for the per-key lease. Migrations are idempotent and run at startup.

`src/infra/schema.sql · src/infra/db.ts`

- [ ] The schema applies twice without error.
- [ ] The outbox has an index on (published_at, id) for the publisher scan.
- [ ] stock_leases has stock_key as its primary key.
- [ ] A test opens an in-memory database and inserts one row in each table.

Risks and decisions

- Retry-After handling is easy to get wrong. Test it with an injected clock, not real sleeps.
- Schema changes after wave 3 force migrations. Get the outbox, version and lease tables right now.

Verification

```
npm run typecheck
npm test -- validation retry db
```

Done when

1. Every acceptance box is checked and the verification commands pass.
2. The session todos for this wave are marked done.
3. The final message includes a hand-off note for the next wave.

Agent brief for this wave: AGENT.md in the same folder.